

- [27] Joakim Nivre, Marie-Catherine de Marneffe, Filip Ginter, Jan Hajic, Christopher D. Manning, Sampo Pyysalo, Sebastian Schuster, Francis Tyers, and Daniel Zeman. Universal Dependencies v2: An evergrowing multilingual treebank collection. In *Proceedings of the 12th Language Resources and Evaluation Conference*, pages 4034–4043, 2020.
- [28] Judea Pearl. *Causality*. Cambridge University Press, Cambridge, UK, 2 edition, 2009. ISBN 978-0-521-89560-6. doi: 10.1017/CBO9780511803161.
- [29] Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.647. URL <https://aclanthology.org/2020.acl-main.647>.
- [30] Shauli Ravfogel, Grusha Prasad, Tal Linzen, and Yoav Goldberg. Counterfactual interventions reveal the causal effect of relative clause representations on agreement prediction. In *Proceedings of the 25th Conference on Computational Natural Language Learning*, pages 194–209, 2021.
- [31] Shauli Ravfogel, Michael Twiton, Yoav Goldberg, and Ryan D Cotterell. Linear adversarial concept erasure. In *International Conference on Machine Learning*, pages 18400–18421. PMLR, 2022.
- [32] Shauli Ravfogel, Francisco Vargas, Yoav Goldberg, and Ryan Cotterell. Adversarial concept erasure in kernel space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 6034–6055, 2022.
- [33] Shauli Ravfogel, Yoav Goldberg, and Ryan Cotterell. Log-linear guardedness and its implications. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9413–9431, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.523. URL <https://aclanthology.org/2023.acl-long.523>.
- [34] Bashir Sadeghi and Vishnu Boddeti. On the fundamental trade-offs in learning invariant representations. *arXiv preprint arXiv:2109.03386*, 2021. URL <https://openreview.net/pdf?id=K0k7mUGspN9>.
- [35] Bashir Sadeghi, Runyi Yu, and Vishnu Boddeti. On the global optima of kernelized adversarial representation learning. In *2019 IEEE/CVF International Conference on Computer Vision*, pages 7970–7978. IEEE, 2019. URL <http://hal.cse.msu.edu/assets/pdfs/papers/2019-iccv-kernel-adversarial-representation-learning.pdf>.
- [36] Shun Shao, Yftah Ziser, and Shay B. Cohen. Gold doesn’t always glitter: Spectral removal of linear and nonlinear guarded attribute information. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1611–1622, Dubrovnik, Croatia, May 2023. Association for Computational Linguistics. URL <https://aclanthology.org/2023.eacl-main.118>.
- [37] Shun Shao, Yftah Ziser, and Shay B. Cohen. Erasure of unaligned attributes from neural representations. *Transactions of the Association for Computational Linguistics*, 11:488–510, 2023. doi: 10.1162/tac1_a_00558. URL <https://aclanthology.org/2023.tac1-1.29>.
- [38] Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, 2019.
- [39] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and efficient foundation language models, 2023. URL <http://arxiv.org/abs/2302.13971>.
- [40] James Francis Woodward. *Making Things Happen: A Theory of Causal Explanation explanation*. Oxford University Press, 2005.

- [41] Zhengxuan Wu, Atticus Geiger, Christopher Potts, and Noah Goodman. Interpretability at scale: Identifying causal mechanisms in Alpaca. 2023.
- [42] Qizhe Xie, Zihang Dai, Yulun Du, Eduard Hovy, and Graham Neubig. Controllable invariance through adversarial feature learning. In *Advances in Neural Information Processing Systems*, volume 30, pages 585–596, 2017. URL <https://dl.acm.org/doi/10.5555/3294771.3294827>.
- [43] Yilun Xu, Shengjia Zhao, Jiaming Song, Russell Stewart, and Stefano Ermon. A theory of usable information under computational constraints. In *8th International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=r1eBeyHFDH>.
- [44] Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, page 335–340, New York, NY, USA, 2018. Association for Computing Machinery. ISBN 9781450360128. URL <https://doi.org/10.1145/3278721.3278779>.

A Additional Related Work

The problem of linear concept erasure is an instance of the general problem of information removal. Information removal methods generally divide into adversarial methods, which are applied during training, and the post-hoc linear methods considered in this paper. Adversarial methods [8, 42, 4, 9, 44] use a gradient-reversal layer during training to induce embeddings that do not encode the protected attribute. However, Elazar and Goldberg [9] have shown that these methods fail in exhaustively removing all the information associated with the protected attribute: it is often possible to train new adversaries that successfully recover the removed information. Linear methods have been proposed as a tractable alternative, where one identifies a linear subspace that captures the concept of interest, and neutralizes it using algebraic techniques. Different methods have been proposed for the identification of the subspace, e.g. PCA and variants thereof [3, 20], orthogonal-rotation [7], classification-based [29], spectral [36, 37] and adversarial approaches [31].

Few works theoretically characterize the condition of linear guardedness. Haghighatkhah et al. [16] extensively analyzed the problem of preventing linear classification, with the focus on decreasing accuracy. They provide a constructive proof of an optimal intervention for an SVM classifier. Ravfogel et al. [33] have proposed a formal definition of linear guardedness based on $\mathcal{V}$ information, and characterized the fairness implications of guardedness; we show the relations with our definition above. Ravfogel et al. [31] provide an adversarial formulation of the problem, derive a closed-form solution to certain cases, and propose an SGD-based optimization for others. While they seek an *orthogonal* projection, we empirically showed that their solution is very close to ours. Sadeghi et al. [35] and Sadeghi and Boddeti [34] both study an adversarial formulation of concept erasure for linear regression, and they trade-off with main-task performance. In contrast to Ravfogel et al. [31], they consider a general linear adversary, i.e. not necessarily a projection matrix. Closest to our work are Kleindessner et al. [20], Haghighatkhah et al. [17], Shao et al. [36]. As we showed above (§ 4), those methods do achieve the goal of linear guardedness though they are unable to prove this fact. At the same time, they are not optimal in terms of damage to the original embedding space.

B Equivalence of Guardedness with the Optimality of Constant Predictors

The following two theorems establish the equivalence of conditions 1 and 2 (indeed, they do so in the general setting, with no assumption of convex loss or linear predictors).

Theorem B.1. *Suppose X ($\mathcal{V}, \mathcal{Z}$)-guards Z . Then for every loss $\mathcal{L} \in \mathcal{L}$, the corresponding trivially attainable loss $L_\tau^{(\mathcal{Z}, \mathcal{L})}$ cannot be improved upon by any predictor $\eta(\cdot; \theta) \in \mathcal{V}$, i.e. $L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$.*

Proof. Consider the null random vector $X'(\omega) = \mathbf{0}$. Since all predictors are constant on X' , and the trivially attainable loss gives the *best* available expected loss among constant predictors, we must have:

$$L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X'; \theta), Z)] \quad (4)$$

The right side of equation (4) is the best possible loss achievable by a function $\eta(\cdot; \theta)$ on the joint distribution of (X', Z) , which by the definition of guardedness is upper bounded by the best possible loss achievable on the joint distribution of (X, Z) :

$$\inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X'; \theta), Z)] \leq \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)] \quad (5)$$

Combining equations (4) and (5), and the fact that all constant functions exist in our function class $\mathcal{V} = \{\eta(\cdot; \theta)\}$, we arrive at our desired result:

$$L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$$

□

Theorem B.2. *Suppose that for every loss $\mathcal{L} \in \mathcal{L}$, the corresponding trivially attainable loss $L_\tau^{(\mathcal{Z}, \mathcal{L})}$ cannot be improved upon by any predictor $\eta(\cdot; \theta) \in \mathcal{V}$, i.e. $L_\tau = \inf_{\theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)]$. Then X ($\mathcal{V}, \mathcal{Z}$)-guards Z .*